

CUSTOMER SHOPPING BEHAVIOUR ANALYSIS

A Data Analytics Project Using Python, SQL, and Power BI

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Tools Used: Python | SQL | Power BI | GitHub

1. Project Overview

This project focuses on analyzing customer shopping behaviour using a dataset containing 3,900 purchase records across different product categories. The objective is to understand customer spending patterns, product preferences, discount usage, subscription behaviour, and repeat purchases.

Python was used to explore, clean, and prepare the dataset. SQL was used to perform SQL-based analysis and answer business questions. Power BI was used to visualize the data and present the analysis in an interactive dashboard.

The analysis aims to identify useful patterns that can help a retail business improve its marketing strategies, product planning, and customer engagement.

Business Problem Statement

A retail company wants to understand its customers' shopping habits to improve sales, customer satisfaction, and long-term loyalty. The company is interested in identifying popular products, understanding the role of discounts and customer reviews, and analyzing purchasing patterns across different customer groups and seasons.

The main objective of this project is to analyze customer shopping data and identify opportunities to improve marketing campaigns, product planning, and customer loyalty.

Key Business Question:

How can the company use customer shopping data to identify trends, improve customer engagement, and develop better marketing and product strategies?

2. Dataset Summary

The dataset contains customer information, purchase details, and shopping behaviour that can be used to understand purchasing patterns.

Dataset Attribute	Description
Total Records	3,900
Total Columns	18
Customer Demographics	Age, Gender, Location, Subscription Status
Purchase Details	Item Purchased, Category, Purchase Amount, Season, Size, Color
Shopping Behaviour	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases
Other Information	Review Rating, Shipping Type, Payment Method
Missing Values	37 values in the Review Rating column

The dataset provides information about customer preferences, purchase amounts, discounts, product categories, payment methods, and previous purchasing behaviour.

Note: The available dataset does not appear to contain a sales-channel column for distinguishing online and offline purchases. Therefore, this project focuses on the fields available in the dataset.

3. Exploratory Data Analysis Using Python

Python was used to explore the dataset, check data quality, and prepare the data for further analysis.

3.1 Data Loading and Initial Exploration

- Imported the dataset into Python using the Pandas library.
- Used `df.info()` to check column names, data types, and non-null values.
- Used `df.describe()` to understand the distribution of numerical columns.
- Reviewed the dataset structure to identify important fields for analysis.

3.2 Data Cleaning and Preparation

- Checked for missing values and duplicate records.
- Handled missing values in the Review Rating column using the median rating of the corresponding product category.
- Renamed columns to snake_case to improve readability and make the column names easier to use in code.
- Created an age_group column to group customers into different age ranges.
- Created a purchase_frequency_days column as part of feature engineering, based on the available purchase-frequency information.
- Checked the relationship between discount_applied and promo_code_used to assess whether the columns contained overlapping information.
- Connected Python to SQL and loaded the prepared dataset for SQL analysis.

```
[3]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                        Non-Null Count  Dtype  
---  --
 0   Customer ID                  3900 non-null   int64   
 1   Age                          3900 non-null   int64   
 2   Gender                       3900 non-null   str      
 3   Item Purchased               3900 non-null   str      
 4   Category                     3900 non-null   str      
 5   Purchase Amount (USD)        3900 non-null   int64   
 6   Location                     3900 non-null   str      
 7   Size                         3900 non-null   str      
 8   Color                        3900 non-null   str      
 9   Season                       3900 non-null   str      
10  Review Rating                3863 non-null   float64  
11  Subscription Status          3900 non-null   str      
12  Shipping Type                3900 non-null   str      
13  Discount Applied             3900 non-null   str      
14  Promo Code Used              3900 non-null   str      
15  Previous Purchases           3900 non-null   int64   
16  Payment Method               3900 non-null   str      
17  Frequency of Purchases       3900 non-null   str      
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

```
[4]: # Summary statistics using .describe()
df.describe(include='all')
```

```
[4]:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

4. Data Analysis Using SQL

SQL was used to analyze the prepared dataset and answer ten business questions related to customer spending, product performance, discounts, subscriptions, and purchasing behaviour.

4.1 Revenue by Gender

Compared the total purchase amount recorded for male and female customers to understand how recorded spending differs between the two groups.

	gender	revenue
1	Male	157890
2	Female	75191

4.2 High-Spending Discount Users

Identified purchases where a discount was applied and the purchase amount was equal to or greater than the overall average purchase amount.

	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97

4.3 Top Five Products by Rating

Identified the five purchased items with the highest average customer review ratings to understand which products received better customer feedback.

	item_purchased	Average Product Rating
1	Gloves	3.860000
2	Sandals	3.840000
3	Boots	3.820000
4	Hat	3.800000
5	Skirt	3.780000

4.4 Shipping Type Comparison

Compared the average purchase amounts for Standard and Express shipping to understand spending patterns across these shipping methods.

	shipping_type	(No column name)
1	Standard	58
2	Express	60

4.5 Subscribers vs. Non-Subscribers

Compared the number of records, average purchase amount, and total recorded purchase amount for subscribed and non-subscribed customers.

	subscription_status	total_customers	avg_spend	total_revenue
1	Yes	1053	59	62645
2	No	2847	59	170436

4.6 Products with High Discount Usage

Identified five products with the highest percentage of purchases where a discount was applied. This analysis helps explore how frequently discounts are associated with purchases of different products.

	item_purchased	discount_rate
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

4.7 Customer Segmentation

Grouped customers into three segments based on their previous-purchase counts:

- **New Customers:** Customers with one previous purchase.
- **Returning Customers:** Customers with two to ten previous purchases.
- **Loyal Customers:** Customers with more than ten previous purchases.

This segmentation helps organize customers according to their recorded purchase history.

	customer_segment	Number of Customers
1	Returning	701
2	Loyal	3116
3	New	83

4.8 Top Three Products per Category

Identified the three most frequently purchased products within each product category to understand product popularity across categories.

	item_rank	category	item_purchased	total_orders
1	1	Accessories	Jewelry	171
2	2	Accessories	Belt	161
3	3	Accessories	Sunglasses	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

4.9 Repeat Buyers and Subscriptions

Analyzed subscription status among records where the previous-purchase count was greater than five to explore the relationship between purchasing history and subscription status.

	subscription_status	repeat_buyers
1	Yes	958
2	No	2518

4.10 Revenue by Age Group

Calculated the total recorded purchase amount for each age group to compare spending contributions across different customer age ranges.

	age_group	total_revenue
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard Development Using Power BI

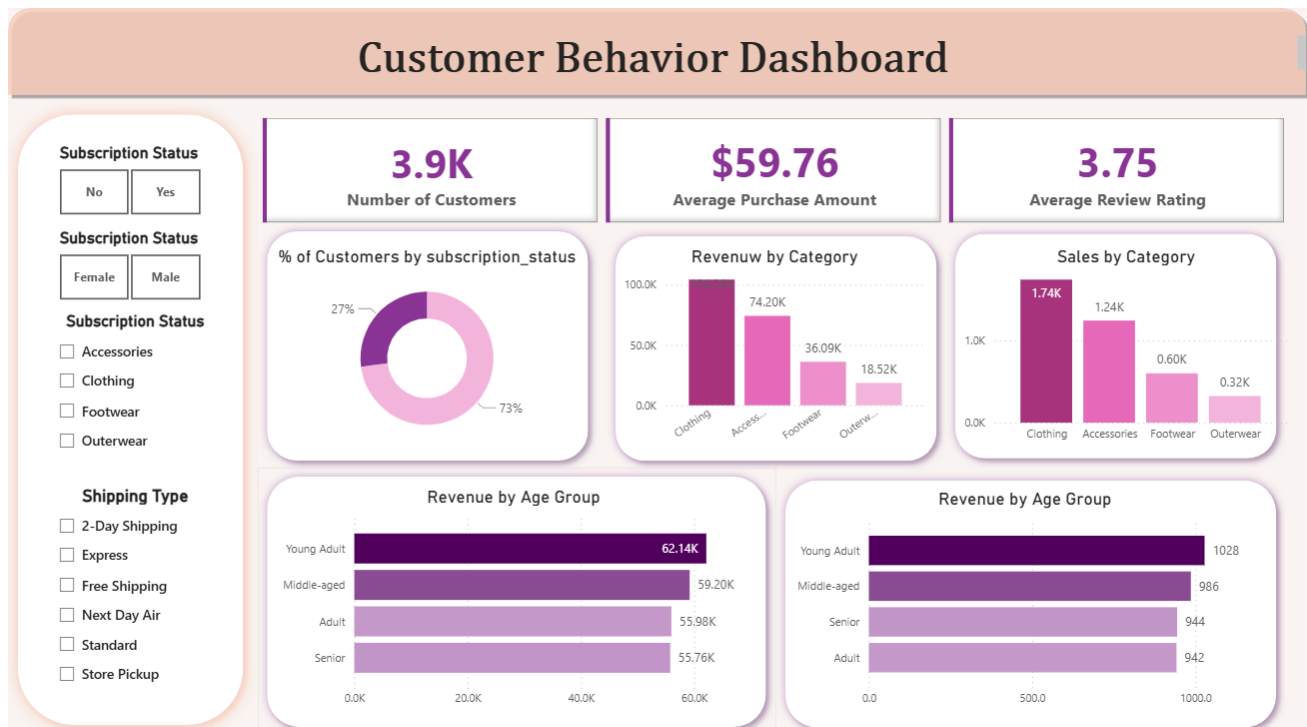
Power BI was used to create an interactive dashboard to present the analysis visually and make the results easier to understand.

The dashboard brings together important aspects of customer shopping behaviour, product preferences, and purchasing patterns.

Key Areas Covered

- **Product Categories:** Compare purchase counts across different categories.
- **Customer Demographics:** Explore shopping behaviour across customer groups.
- **Discount Usage:** Examine purchases associated with discounts and promotional codes.
- **Payment Preferences:** Understand the distribution of payment methods.
- **Customer Ratings:** Compare average review ratings across relevant product or shipping groups.
- **Seasonal Patterns:** Explore differences in purchasing behaviour across seasons.
- **Subscription Status:** Compare purchasing patterns between subscribed and non-subscribed customers.
- **Customer Purchase History:** Explore purchasing behaviour using previous-purchase information.

Charts, filters, and slicers allow users to interact with the dashboard and examine different parts of the dataset.



6. Tools and Technologies Used

Tool	Purpose
Python	Data exploration, cleaning, and preparation
Pandas	Data manipulation and analysis
SQL/ SSMS	Data storage and SQL-based analysis
Power BI	Interactive dashboard development and visualization
GitHub	Project version control and documentation

7. Business Recommendations

Based on the analysis areas covered in this project, the following actions could be considered by the retail business. The effectiveness of each recommendation should be evaluated using the actual analysis results.

7.1 Encourage Subscriptions

Explore subscription benefits, exclusive offers, and personalized promotions to encourage customers to subscribe. Compare purchase behaviour before and after implementing these initiatives.

7.2 Develop Customer Loyalty Programs

Consider reward points, repeat-purchase offers, and personalized benefits to encourage returning customers to continue shopping with the business.

7.3 Review Discount Policies

Analyze discount usage alongside purchase amounts. Where profit-margin data is available, use it to assess whether discounts are generating sufficient business value.

7.4 Improve Product Positioning

Use highly rated and frequently purchased products to guide product recommendations, promotional campaigns, and product placement.

7.5 Use Targeted Marketing

Use customer demographics and purchasing patterns to develop more relevant marketing campaigns. Measure campaign performance to determine which customer groups respond best.

7.6 Improve Seasonal Planning

Review purchasing patterns across seasons to help plan promotional activities and product availability according to customer preferences.

8. Conclusion

This project demonstrates how Python, SQL, and Power BI can be used together to analyze customer shopping behaviour.

Python was used to explore, clean, and prepare the dataset. SQL was used to answer business questions related to spending patterns, product ratings, discount usage, customer segments, and subscription behaviour. Power BI was used to present the analysis in an interactive dashboard.

The project provides a structured approach to understanding customer preferences and purchasing patterns. The findings can support decisions related to marketing, product planning, promotional strategies, and customer engagement.

Further analysis using additional information, such as profit margins, transaction dates, and sales channels, could provide a more complete understanding of retail performance.

End of Report